Milin Doppalapudi

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Education

Master of Science, Computer Science — Sept. 2024 to Dec. 2025
University of California, Santa Cruz
- GPA: 4.00 (Transcript)

Bachelor of Science, Computer Science — Sept. 2020 to May 2024
University of California, Santa Cruz
- GPA: 3.96 (Transcript)

Work Experience

NTT Research PHI Lab, Intern — CA, USA — June 2024 to December 2024
- Research intern studying application of solving the quadratic assignment problem (QAP) using graph neural networks and the experimentation of GNNs for SAT solver selection.

University of California Santa Cruz, Tutor / Grader — CA, USA — Sep 2022 to Mar 2024
- Assisted computer science students on specific projects, closely reviewing materials and assignments to offer targeted help

Projects

Live Text Detector
- Developed two blur-robust real-time live video text detection models using DBNet and DINOv2, achieving over 50% relative F1 improvement over the base DBNet model under heavy motion blur.

Expression Recognition Deep Learning Tic-Tac-Toe Game
- Developed a facial expression recognition model to classify emotions (happy, sad, neutral) and integrated it into an interactive Tic-Tac-Toe game that uses real-time user expressions to control gameplay.

PEFT LLaVa-Med for X-Ray Disease Classification
- Finetuned the multimodal LLaVA-Med model using parameter-efficient methods to detect pneumothorax and multiclass diseases from chest X-rays, achieving a 25%+ absolute improvement in F1-score over zero-shot baselines on the NIH ChestX-ray14 dataset.

GNN SAT Solver Selector
- Designed and trained a graph neural network to predict the optimal SAT solver for a given Boolean satisfiability instance, optimizing solver selection efficiency across large CNF datasets.

Skills

Languages
- Python (6 Yrs), C (5 Yrs), C++ (5 Yrs), SQL (2 Yrs)

Machine Learning / Deep Learning
- PyTorch, TensorFlow, HuggingFace Transformers, scikit-learn, NumPy, SciPy, pandas, etc

Computer Vision
- OpenCV, torchvision, PIL, image augmentation, detection & segmentation

Systems / Tools
- AWS, Linux, Git, LaTeX, Docker

Publications

- Jiang, E., Leleu, T., Reifenstein, S., & Doppalapudi, M. “Graph Neural Networks for Hyperparameter Inference in Ising Solvers.” Optimization for ML (OPT) Workshop at NeurIPS 2024, 2024.